

Internet of Things and Wireless Networks

Lab 3, Part 2, 3 – Renode Instructions

Communication between Renode simulation and host

```
emulation CreateSwitch "switch"
```

```
mach create "publisher"
```

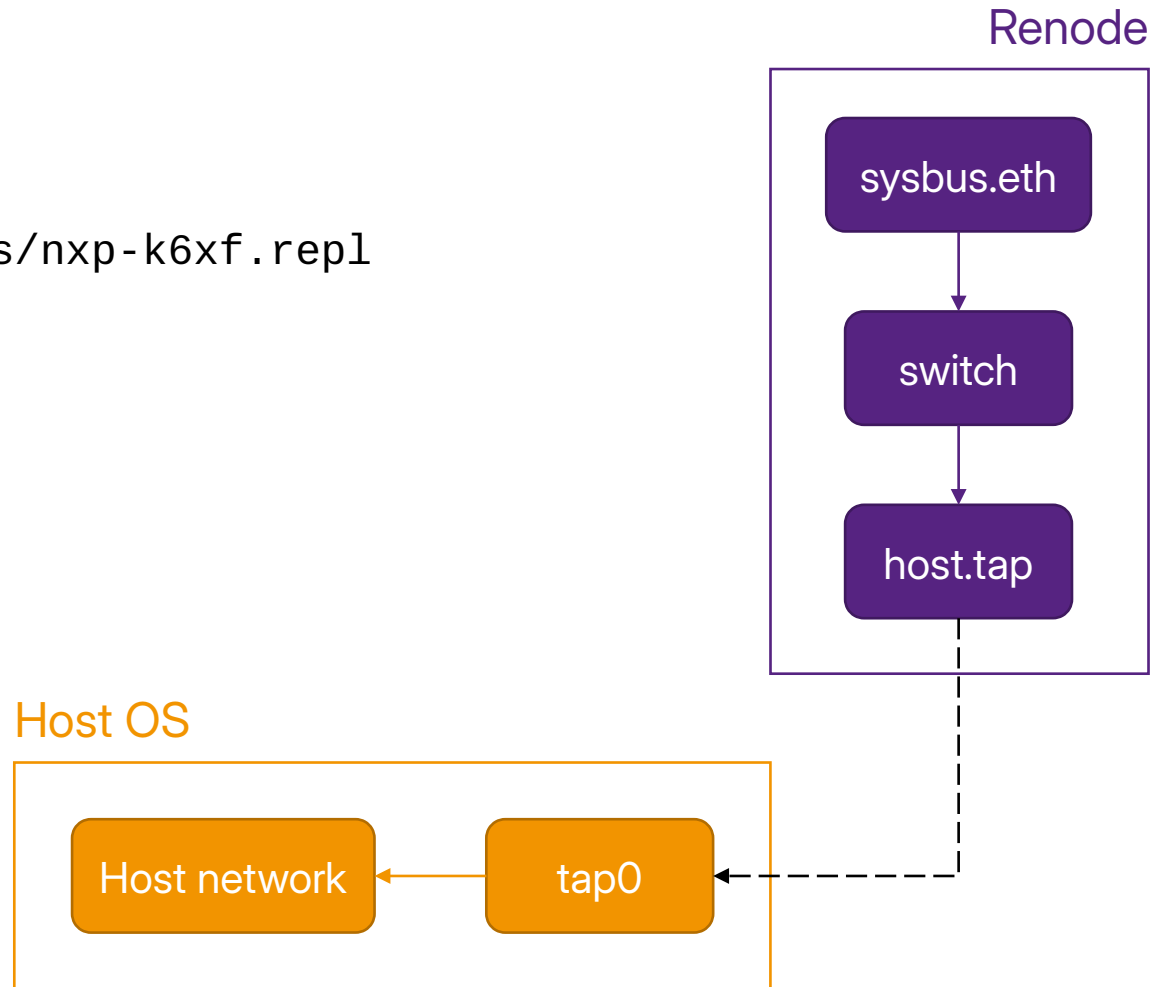
```
machine LoadPlatformDescription @platforms/cpus/nxp-k6xf.repl
```

```
connector Connect sysbus.eth switch
```

```
emulation CreateTap "tap0" "tap"
```

```
connector Connect host.tap switch
```

Create tap0 on your host OS before starting the Renode simulation.



Creating a TAP interface on Ubuntu

Execute these commands in the terminal:

```
sudo ip tuntap add dev tap0 mode tap  
sudo ip addr add 192.0.2.2/24 dev tap0  
sudo ip link set dev tap0 up
```

Creating a TAP interface on Windows

Documented in Renode documentation: <https://renode.readthedocs.io/en/latest/networking/host-network.html>

TAP networking seems to be harder to set up on Windows. Last year, we couldn't get this setup to work although it should have. Do not spend too much time on this if you still want to try it. It is way easier on Linux. Or, of course, use the real hardware we provide in HRS 3, room 502.

Configure Zephyr (static IP)

```
# Networking core
```

```
CONFIG_NETWORKING=y
```

```
CONFIG_NET_IPV4=y
```

```
CONFIG_NET_TCP=y
```

```
CONFIG_NET_SOCKETS=y
```

```
# Ethernet driver
```

```
CONFIG_NET_L2_ETHERNET=y
```

```
# For FRDM-K64F; use the right driver for your  
board
```

```
CONFIG_ETH_MCUX=y
```

You can of course configure IPv6 here as well.

```
# Static IPv4 configuration
```

```
CONFIG_NET_CONFIG_SETTINGS=y
```

```
CONFIG_NET_CONFIG_NEED_IPV4=y
```

```
CONFIG_NET_CONFIG_MY_IPV4_ADDR="192.0.2.1"
```

```
CONFIG_NET_CONFIG_MY_IPV4_NETMASK="255.255.255.0"
```

```
CONFIG_NET_CONFIG_MY_IPV4_GW="192.0.2.2"
```

```
# MQTT
```

```
CONFIG_MQTT_LIB=y
```